

Pitch for experimentally-constrained multi-region RNN modelling of Hippocampal-Cortical Interactions

The hippocampus is widely recognized as the brain's hub for forming and retrieving episodic memories, binding together sensory details into coherent event representations. In concert with neocortical regions, it encodes the spatial and contextual scaffolds of experience, then later reactivates these patterns during consolidation. These hippocampal–cortical interactions – especially those accompanying sharp-wave ripples (SWRs) – are thought to underlie memory replay and long-term storage. For example, prefrontal and hippocampal circuits have distinct but complementary roles in episodic memory, and they interact via oscillatory coupling to retrieve and integrate contextual information pubmed.ncbi.nlm.nih.gov. Similarly, recent imaging studies show that hippocampal SWRs drive activity in default-mode cortical networks (e.g. retrosplenial cortex), implying bidirectional signaling during sleep that could mediate consolidation pubmed.ncbi.nlm.nih.gov. Together, these findings emphasize that hippocampal-cortical loops are crucial for navigating and remembering spatial environments and for replaying experiences.

Building on this biology, recent work has developed multi-region recurrent neural network (RNN) models that are tightly constrained by experimental data. Rajan and colleagues introduced a framework in which RNNs are trained directly on time-series recordings from multiple brain areas; once trained, the model's recurrent weight matrices can be decomposed to infer directed interactions and input currents among regions imagwiki.nibib.nih.gov. In the *Current-based Decomposition* (CURBD) approach, an RNN is fit to recorded neural activity (e.g. calcium or spikes) so that its connectivity reproduces the data; one then examines the model's effective weight matrix to compute the excitatory and inhibitory drives from each source onto each target neuron. This “de-mixing” of current inputs across regions reveals how signals propagate in vivo in ways that raw data alone cannot show imagwiki.nibib.nih.gov. In parallel, the TRAKR framework uses RNNs to detect latent state transitions in neural and behavioral recordings, flagging moments when the underlying circuit shifts between modes (e.g. task engagement vs. rest) imagwiki.nibib.nih.gov. Both tools exemplify how theory-driven RNN models, grounded in data, can reverse-engineer multi-area circuit dynamics. By applying CURBD to simultaneous hippocampus-cortex recordings, for instance, one can in principle map the directed influence of hippocampal outputs on cortical targets during replay, or vice versa, yielding a detailed model of inter-regional coordination that is otherwise hidden. In short, these multi-region RNN methods offer a powerful way to “trace” functional connectivity and circuit motifs in the intact brain imagwiki.nibib.nih.gov imagwiki.nibib.nih.gov.

Meanwhile, theoretical models of hippocampal cognitive maps have been advancing our understanding of how the hippocampus encodes space and relational structure. Whittington *et al.* (2020) proposed the Tolman–Eichenbaum Machine (TEM), a neural model in which hippocampal and entorhinal representations are factorized into separate components. In TEM, structural knowledge (the graph or spatial layout of an environment) is encoded in one set of neural codes, while sensory/contextual features (landmarks, cues) are encoded in another. These are combined conjunctively in a Hebbian memory system so that the animal “knows what

was where”pubmed.ncbi.nlm.nih.gov. This factorization allows generalization: the model can transfer structural knowledge across different sensory environments and predict novel trajectories. Importantly, when TEM is trained on simulated spatial or non-spatial tasks, the learned units naturally resemble biological place cells, grid cells, and other hippocampal–entorhinal responsespubmed.ncbi.nlm.nih.gov. In other work, Whittington and colleagues showed that artificial Transformer networks – standard architectures in machine learning – can also recapitulate hippocampal spatial codes. By adding a recurrent positional encoding, a Transformer trained on sequences reproduces precise place and grid cell firing patterns, effectively acting like an abstracted hippocampal modelopenreview.net. Notably, this Transformer-based approach yielded performance gains over prior neuroscience models, suggesting a deep algorithmic link between artificial sequence models and brain navigation systems. Taken together, Whittington’s work unifies spatial and non-spatial cognitive maps: hippocampal circuits factorize environmental structure versus content and could even operate with computations analogous to modern deep learning architectures (e.g. Transformers) to encode and replay experiencespubmed.ncbi.nlm.nih.govopenreview.net.

In parallel, methodological advances have shown how RNNs themselves can serve as powerful surrogates for neural circuits. Durstewitz *et al.* (2023) review how RNNs trained on neural or behavioral data effectively reconstruct the underlying dynamical systempubmed.ncbi.nlm.nih.gov. Because any autonomous neural circuit can be seen as a system of differential equations, one can in principle train an RNN to emulate those dynamics. Once the RNN is fit to data, it becomes a “formal surrogate” that inherits the temporal and geometric properties of the biological systempubmed.ncbi.nlm.nih.gov. This means one can analyze, perturb, or simulate the model to probe mechanisms inaccessible in vivo. Durstewitz *et al.* dub this process “dynamical system reconstruction” and emphasize its promise for neurosciencepubmed.ncbi.nlm.nih.gov. In practice, one might fit an RNN to recordings from hippocampus and cortex during a memory task; the trained RNN would encode the latent dynamics of both regions and their coupling. By examining this model (for instance via CURBD), one could then infer how hippocampal sequences drive cortical patterns or which network states correspond to replay. Thus, combining theory-based models like TEM and data-driven RNN surrogates offers a rich framework: the former provides normative principles of memory coding, while the latter yields an empirical dynamical portrait of actual circuitspubmed.ncbi.nlm.nih.govpubmed.ncbi.nlm.nih.gov.

This perspective is echoed in recent reviews of navigation neuroscience. As discussed by Spiers *et al.* (2023), navigating space engages a distributed network: the hippocampus and entorhinal cortex encode the animal’s location and the paths it has taken, while cortical areas (such as retrosplenial, parietal, and frontal cortices) contribute goals, decision variables, and extended context. These hippocampal–cortical loops enable flexible planning and memory-guided behavior. For example, hippocampal spatial codes (place/grid cells) may propagate through retrosplenial and prefrontal areas to compute future trajectories and action plans. In turn, cortical feedback may modulate hippocampal encoding based on goal relevance. Empirical studies support this: Pedrosa *et al.* (2022) found that neocortical activity – especially in a default-mode network centered on retrosplenial cortex – both precedes and follows hippocampal SWRspubmed.ncbi.nlm.nih.gov. Moreover, they observed that hippocampal

slow-gamma rhythms (often associated with CA3 inputs) were strongly linked to cortical activation transients, suggesting bidirectional exchange during sleep pubmed.ncbi.nlm.nih.gov. Altogether, these findings underline the idea that cognition during navigation relies on coordinated computations across hippocampus and cortex: memory for places and routes is distributed, with the hippocampus anchoring spatial representations and cortex integrating them into broader cognitive maps.

Importantly, all of the above suggests that **multi-region RNN modeling** – grounded in both theory and data – can be used to simulate hippocampal–cortical interactions during memory replay. By training an RNN on simultaneous recordings from hippocampus and connected cortical regions (for example, wide-field calcium imaging of cortex together with hippocampal LFP/spikes during a spatial task and rest), one would obtain a model whose dynamics mirror the real system. The CURBD analysis of such an RNN could then reveal the directed coupling during SWRs: it might show, for instance, how a hippocampal-generated sequence (a replay event) propagates excitation to retrosplenial or visual cortex, and conversely how cortical input (such as a prefrontal signal) shapes hippocampal timing. Because the RNN is constrained by data, its inferred connection strengths and currents would reflect the biology. Meanwhile, the trained model itself could be “replayed” in silico: one could trigger synthetic SWR-like patterns and observe the resultant cortical activity, testing hypotheses about timing and spread. This approach effectively implements the dynamical system reconstruction idea pubmed.ncbi.nlm.nih.gov. Additionally, one could apply TRAKR to the real data to identify transitions (e.g. awake to SWR states) and use the RNN to predict where those transitions originate and terminate. Thus, a multi-area RNN provides a bridge between computational memory models (like TEM) and experiment: it can simulate hippocampal-cortical replay loops under realistic constraints, guiding interpretations of observed LFP and imaging signals.

Finally, a concrete project could unite these computational and experimental threads in Dr. Mohajerani’s lab. For example, one might train mice on a spatial learning task while recording both hippocampal electrophysiology (to capture spikes and ripples) and wide-field cortical calcium imaging (to map cortex activity). Using this dataset, a computational team would train a multi-region RNN (CURBD-based) that takes hippocampal and cortical activity as inputs. The model’s inferred weights would predict how hippocampal ripples drive specific cortical networks (e.g. retrosplenial and visual areas) and vice versa. To test this, the experimental arm could optogenetically perturb predicted nodes: for instance, silencing retrosplenial cortex during detected SWRs to see if hippocampal replay patterns are altered as the model predicts. In addition, TRAKR could be used to mark putative “replay state” onsets in the data, which can be compared against the RNN’s internal state transitions. By iterating between model predictions and targeted interventions, one would both validate the RNN’s dynamical reconstruction and uncover causal roles of hippocampal–cortical loops in memory replay. This synergy – RNN-guided hypotheses driving new experiments – would be a natural fit for Dr. Mohajerani’s focus on hippocampal-cortical replay coordination, blending cutting-edge computation with in vivo circuit analysis.